

E-COMMERCE SALES & PROFITABILITY ANALYSIS

Complete End-to-End Data Analysis Project

1. PROJECT OVERVIEW

1.1 Project Title

E-Commerce Sales and Profitability Analysis

1.2 Project Type

Data Analysis / Business Intelligence / Sales Analytics

1.3 Dataset Size

3,500 historical sales records

1.4 Industry

E-Commerce / Retail

1.5 Project Objective

The objective of this project is to analyze e-commerce sales data to identify **business trends, product performance, regional demand, and profitability patterns**.

The analysis will help management understand which products and regions generate the highest sales and profit, identify products with high sales but low profitability, and develop data-driven recommendations for improving business performance.

2. BUSINESS PROBLEM

2.1 Problem Statement

An e-commerce company operates across four regions and sells products from three major categories:

- Technology
- Furniture
- Office Supplies

Although the company has historical sales data, management needs a better understanding of:

- Sales performance over time
- Regional demand
- Product performance
- Revenue generation
- Profitability
- Products with high sales but low profit

The company wants to use data analysis to improve product strategies, pricing decisions, and regional performance.

3. PROJECT OBJECTIVES

The project will focus on the following objectives.

3.1 Sales Performance Analysis

- Calculate total sales.
- Calculate total profit.
- Identify the number of transactions.
- Analyze monthly sales trends.
- Identify the highest and lowest sales months.

3.2 Product Performance Analysis

- Identify top-selling products.
- Analyze sales by product category.
- Identify products contributing the most revenue.
- Compare product quantity and sales.
- Identify low-performing products.

3.3 Regional Analysis

- Compare sales across four regions.
- Identify the highest-performing region.
- Analyze regional profit.
- Compare product-category performance across regions.

3.4 Profitability Analysis

- Calculate profit margins.
- Identify high-sales, low-profit products.
- Compare sales and profit by category.
- Identify products that may require pricing or cost reviews.

3.5 Business Recommendations

- Recommend strategies to improve profitability.
 - Identify opportunities in high-performing regions.
 - Suggest product-level pricing or cost investigations.
 - Support management decisions using data-driven insights.
-

4. DATASET UNDERSTANDING

4.1 Dataset Description

The dataset contains **3,500 historical sales records** from a simulated e-commerce company.

Each record represents a sales transaction containing information about:

- Order date
- Product
- Product category
- Region
- Quantity
- Sales
- Profit

4.2 Dataset Columns

Column	Description	Data Type
Order Date	Date on which the transaction occurred	Date

Product Name	Name of the product	Categorical
Category	Product category	Categorical
Region	Sales region	Categorical
Quantity	Number of units sold	Numerical
Sales	Revenue generated from the transaction	Numerical
Profit	Profit generated from the transaction	Numerical

Note: The exact column names should be confirmed after loading the actual dataset.

5. BUSINESS QUESTIONS

Before starting the analysis, convert the business problem into measurable questions.

5.1 General Sales Questions

1. What is the total sales revenue?
2. What is the total profit?
3. How many transactions are recorded?
4. What is the average transaction value?
5. What is the total quantity sold?

5.2 Time-Based Questions

1. How do sales change month by month?
2. Which month generated the highest revenue?
3. Which month generated the highest profit?
4. Are there seasonal sales patterns?
5. Are sales increasing or decreasing over time?

5.3 Product Questions

1. Which products generate the highest sales?
2. Which products generate the highest profit?
3. Which products sell the highest quantity?
4. Which categories perform best?
5. Which products have high sales but low profit?

5.4 Regional Questions

1. Which region generates the highest sales?
2. Which region generates the highest profit?
3. Which region sells the most products?
4. Which categories perform best in each region?
5. Which regions have low profitability?

5.5 Profitability Questions

1. What is the overall profit margin?
 2. Which products have the highest profit margins?
 3. Which products have high revenue but low profit?
 4. Which categories generate high sales but comparatively low profit?
 5. Where should management investigate pricing or costs?
-

6. PROJECT SCOPE

6.1 In Scope

This project will include:

- Data loading
- Data profiling
- Data cleaning
- Data validation
- Feature engineering
- Exploratory Data Analysis
- Time-series analysis
- Product analysis
- Category analysis
- Regional analysis
- Profitability analysis
- SQL analysis
- Power BI dashboard development
- KPI creation
- Business insights
- Recommendations

6.2 Out of Scope

This project will not include:

- Machine learning
- Sales forecasting
- Customer-level analysis, unless customer data is available
- Inventory optimization, unless inventory data is available
- Real-time sales monitoring
- Automated pricing systems

This keeps the project focused on **Data Analysis and Business Intelligence**.

7. PROJECT TECHNOLOGY STACK

7.1 Python

Used for:

- Data loading
- Data cleaning
- Data transformation
- Exploratory Data Analysis
- Statistical analysis
- Visualization

Libraries

- Pandas
- NumPy
- Matplotlib
- Seaborn

7.2 SQL

Used for:

- Data querying
- Aggregation
- Grouping
- Product analysis
- Regional analysis
- Profitability analysis

Database

- MySQL

7.3 Power BI

Used for:

- Data modeling
- KPI creation
- Interactive dashboards
- Business reporting
- Data visualization

7.4 Documentation

- Jupyter Notebook
 - GitHub
 - Project Report
 - Power BI `.pbix` file
-

8. PROJECT WORKFLOW

The project will follow this process:

Business Problem



Define Objectives



Understand Dataset



Load Data



Data Profiling



Data Cleaning



Data Transformation



Feature Engineering



Exploratory Data Analysis



Sales Analysis



Product Analysis



Regional Analysis
↓
Profitability Analysis
↓
SQL Analysis
↓
Power BI Dashboard
↓
Key Insights
↓
Business Recommendations
↓
Validation
↓
Documentation
↓
GitHub
↓
Resume & Interview Preparation

9. DATA COLLECTION

9.1 Dataset File

For example:

ecommerce_sales.csv

9.2 Project Folder

E-Commerce-Sales-Analysis/

```
|
|— data/
|   └─ ecommerce_sales.csv
|
|— notebooks/
|   └─ ecommerce_analysis.ipynb
|
|— sql/
|   └─ analysis_queries.sql
|
|— powerbi/
|   └─ ecommerce_dashboard.pbix
|
```

```
|— reports/
|   |— project_report.pdf
|
|— README.md
```

10. DATA LOADING USING PYTHON

10.1 Import Libraries

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

10.2 Load Dataset

```
df = pd.read_csv("ecommerce_sales.csv")
```

10.3 Display First Five Rows

```
df.head()
```

10.4 Check Dataset Dimensions

```
df.shape
```

Expected number of rows:

3,500

The number of columns should be confirmed from the actual file.

11. DATA UNDERSTANDING

11.1 View Column Names

```
df.columns
```

11.2 Check Data Types

```
df.info()
```

Identify:

- Date columns
- Numeric columns
- Categorical columns

11.3 Generate Descriptive Statistics

```
df.describe()
```

This helps understand:

- Average sales
 - Minimum sales
 - Maximum sales
 - Average quantity
 - Profit distribution
-

12. DATA PROFILING

Data profiling helps identify data-quality issues before analysis.

12.1 Missing Value Analysis

```
df.isnull().sum()
```

Questions:

- Are any order dates missing?
- Are product names missing?
- Are sales values missing?
- Are profit values missing?

12.2 Duplicate Analysis

```
df.duplicated().sum()
```

Investigate duplicate transactions before removing them.

12.3 Unique Values

```
df["Category"].unique()
```

```
df["Region"].unique()
```

Check whether the dataset contains the expected categories and regions.

13. DATA CLEANING

13.1 Handle Missing Values

Depending on the business meaning of the column:

- Remove invalid records.
- Replace missing categorical values with "Unknown".
- Investigate missing financial values.
- Avoid replacing missing sales or profit with zero without justification.

13.2 Remove Duplicate Records

If a record is confirmed to be a duplicate:

```
df = df.drop_duplicates()
```

13.3 Convert Order Date

```
df["Order Date"] = pd.to_datetime(df["Order Date"])
```

13.4 Convert Numeric Columns

```
df["Sales"] = pd.to_numeric(df["Sales"])
df["Profit"] = pd.to_numeric(df["Profit"])
df["Quantity"] = pd.to_numeric(df["Quantity"])
```

Use the actual column names in your dataset.

14. DATA VALIDATION

Check for incorrect or unrealistic values.

14.1 Quantity Validation

Investigate:

- Negative quantities
- Zero quantities

- Unusually large quantities

```
df[df["Quantity"] <= 0]
```

14.2 Sales Validation

Investigate:

- Negative sales
- Zero sales
- Extremely high sales

```
df[df["Sales"] < 0]
```

14.3 Profit Validation

Negative profit may be legitimate if a transaction resulted in a loss.

Therefore, don't automatically delete negative-profit records.

```
df[df["Profit"] < 0]
```

15. FEATURE ENGINEERING

Create new columns that help answer business questions.

15.1 Year

```
df["Year"] = df["Order Date"].dt.year
```

15.2 Month

```
df["Month"] = df["Order Date"].dt.month
```

15.3 Month Name

```
df["Month_Name"] = df["Order Date"].dt.month_name()
```

15.4 Quarter

```
df["Quarter"] = df["Order Date"].dt.quarter
```

15.5 Profit Margin

If sales are greater than zero:

```
df["Profit_Margin"] = (  
    df["Profit"] / df["Sales"]  
) * 100
```

Important

The transaction-level profit margin should not automatically be averaged to calculate the overall company margin.

The overall margin should be calculated as:

Total Profit / Total Sales × 100

16. EXPLORATORY DATA ANALYSIS — EDA

EDA is the process of exploring the dataset to discover patterns, relationships, and unusual behavior.

The analysis will cover:

1. Overall sales performance
 2. Time-based trends
 3. Product performance
 4. Category performance
 5. Regional performance
 6. Profitability
-

17. OVERALL SALES ANALYSIS

17.1 Total Sales

```
total_sales = df["Sales"].sum()
```

17.2 Total Profit

```
total_profit = df["Profit"].sum()
```

17.3 Total Quantity Sold

```
total_quantity = df["Quantity"].sum()
```

17.4 Total Transactions

```
total_transactions = len(df)
```

If the dataset contains a unique order ID, use the appropriate distinct-order calculation instead.

17.5 Average Transaction Value

```
average_transaction_value = (  
    total_sales / total_transactions  
)
```

18. MONTHLY SALES TREND ANALYSIS

18.1 Objective

Identify monthly fluctuations and seasonal sales patterns.

18.2 Calculate Monthly Sales

```
monthly_sales = (  
    df.groupby("Month")["Sales"]  
        .sum()  
        .reset_index()  
)
```

18.3 Visualize Monthly Sales

```
plt.figure(figsize=(10, 5))
```

```
sns.lineplot(  
    data=monthly_sales,  
    x="Month",  
    y="Sales",  
    marker="o"  
)
```

```
plt.title("Monthly Sales Trend")  
plt.xlabel("Month")
```

```
plt.ylabel("Total Sales")
plt.show()
```

18.4 Business Questions

- Which month generated the highest sales?
 - Which month generated the lowest sales?
 - Are there seasonal patterns?
 - Are sales increasing over time?
-

19. MONTHLY PROFIT ANALYSIS

Calculate profit by month.

```
monthly_profit = (
    df.groupby("Month")["Profit"]
    .sum()
    .reset_index()
)
```

Compare monthly sales and profit.

Key Question

Does the month with the highest sales also generate the highest profit?

This is important because high revenue does not always mean high profitability.

20. PRODUCT PERFORMANCE ANALYSIS

20.1 Sales by Product

```
product_sales = (
    df.groupby("Product Name")["Sales"]
    .sum()
    .sort_values(ascending=False)
)
```

20.2 Top 10 Products by Sales

```
product_sales.head(10)
```

20.3 Top 10 Products by Profit

```
product_profit = (  
    df.groupby("Product Name")["Profit"]  
        .sum()  
        .sort_values(ascending=False)  
)
```

```
product_profit.head(10)
```

20.4 Top Products by Quantity

```
product_quantity = (  
    df.groupby("Product Name")["Quantity"]  
        .sum()  
        .sort_values(ascending=False)  
)
```

```
product_quantity.head(10)
```

21. CATEGORY ANALYSIS

The dataset contains three main categories:

- Technology
- Furniture
- Office Supplies

21.1 Sales by Category

```
category_sales = (  
    df.groupby("Category")["Sales"]  
        .sum()  
        .sort_values(ascending=False)  
)
```

21.2 Profit by Category

```
category_profit = (  
    df.groupby("Category")["Profit"]  
        .sum()  
)
```

```
.sort_values(ascending=False)
)
```

21.3 Quantity by Category

```
category_quantity = (
    df.groupby("Category")["Quantity"]
    .sum()
    .sort_values(ascending=False)
)
```

21.4 Business Questions

- Which category generates the highest revenue?
 - Which category generates the highest profit?
 - Which category sells the highest quantity?
 - Does the highest-sales category also have the highest profit?
-

22. REGIONAL ANALYSIS

The dataset contains four regions.

22.1 Sales by Region

```
region_sales = (
    df.groupby("Region")["Sales"]
    .sum()
    .sort_values(ascending=False)
)
```

22.2 Profit by Region

```
region_profit = (
    df.groupby("Region")["Profit"]
    .sum()
    .sort_values(ascending=False)
)
```

22.3 Quantity by Region

```
region_quantity = (
    df.groupby("Region")["Quantity"]
    .sum()
)
```

```
.sort_values(ascending=False)
)
```

22.4 Business Questions

- Which region generates the highest sales?
 - Which region generates the highest profit?
 - Which region sells the most units?
 - Are some regions generating high sales but low profit?
-

23. REGIONAL CATEGORY ANALYSIS

Analyze category performance within each region.

```
regional_category_sales = (
    df.groupby(["Region", "Category"])["Sales"]
      .sum()
      .reset_index()
)
```

This can answer:

Which product categories perform best in each region?

Example analysis structure:

Region	Category	Sales	Profit
Region A	Technology	—	—
Region A	Furniture	—	—
Region A	Office Supplies	—	—

Populate the results using actual calculations.

24. PROFITABILITY ANALYSIS

This is one of the most important sections of the project.

24.1 Overall Profit Margin

```
overall_profit_margin = (  
    df["Profit"].sum() / df["Sales"].sum()  
) * 100
```

24.2 Profit Margin by Category

```
category_summary = (  
    df.groupby("Category")  
    .agg(  
        Total_Sales=("Sales", "sum"),  
        Total_Profit=("Profit", "sum")  
    )  
    .reset_index()  
)
```

```
category_summary["Profit_Margin"] = (  
    category_summary["Total_Profit"] /  
    category_summary["Total_Sales"]  
) * 100
```

24.3 Profit Margin by Region

```
region_summary = (  
    df.groupby("Region")  
    .agg(  
        Total_Sales=("Sales", "sum"),  
        Total_Profit=("Profit", "sum")  
    )  
    .reset_index()  
)
```

```
region_summary["Profit_Margin"] = (  
    region_summary["Total_Profit"] /  
    region_summary["Total_Sales"]  
) * 100
```

25. HIGH SALES, LOW PROFIT ANALYSIS

25.1 Business Problem

Some products may generate substantial revenue but contribute relatively little profit.

For example:

Product A
High Sales
Low Profit Margin

This may require further investigation into:

- Pricing
- Discounts
- Product costs
- Shipping expenses
- Other transaction-level costs, if available

25.2 Product-Level Summary

```
product_summary = (  
    df.groupby("Product Name")  
    .agg(  
        Total_Sales=("Sales", "sum"),  
        Total_Profit=("Profit", "sum"),  
        Total_Quantity=("Quantity", "sum")  
    )  
    .reset_index()  
)
```

```
product_summary["Profit_Margin"] = (  
    product_summary["Total_Profit"] /  
    product_summary["Total_Sales"]  
) * 100
```

25.3 Identify High-Sales, Low-Profit Products

You must define what **high sales** and **low profit** mean.

For example:

- High sales: Above the median product sales
- Low profit margin: Below the median product profit margin

```
high_sales_threshold = product_summary["Total_Sales"].median()  
low_margin_threshold = product_summary["Profit_Margin"].median()
```

```
high_sales_low_profit = product_summary[  
    (product_summary["Total_Sales"] > high_sales_threshold) &  
    (product_summary["Profit_Margin"] < low_margin_threshold)
```

]

Important

The median is only one possible method. You can also use business-defined thresholds or percentiles.

26. SALES VS PROFIT ANALYSIS

Create a scatter plot comparing product sales and profit.

```
plt.figure(figsize=(10, 6))

sns.scatterplot(
    data=product_summary,
    x="Total_Sales",
    y="Total_Profit"
)

plt.title("Product Sales vs Profit")
plt.xlabel("Total Sales")
plt.ylabel("Total Profit")
plt.show()
```

Questions

- Which products generate high sales and high profit?
 - Which products generate high sales but low profit?
 - Which products generate low sales but high profit margins?
-

27. QUANTITY VS PROFIT ANALYSIS

Analyze whether products selling more units necessarily generate more profit.

```
plt.figure(figsize=(10, 6))

sns.scatterplot(
    data=product_summary,
    x="Total_Quantity",
    y="Total_Profit"
)
```

```
plt.title("Quantity Sold vs Profit")
plt.xlabel("Total Quantity")
plt.ylabel("Total Profit")
plt.show()
```

28. STATISTICAL ANALYSIS

Use descriptive statistics to understand the financial data.

28.1 Sales Statistics

```
df["Sales"].describe()
```

28.2 Profit Statistics

```
df["Profit"].describe()
```

28.3 Quantity Statistics

```
df["Quantity"].describe()
```

Analyze:

- Mean
 - Median
 - Standard deviation
 - Minimum
 - Maximum
 - Quartiles
-

29. CORRELATION ANALYSIS

Analyze relationships between numerical variables.

```
df[["Quantity", "Sales", "Profit"]].corr()
```

Visualize:

```
sns.heatmap(
    df[["Quantity", "Sales", "Profit"]].corr(),
    annot=True
```

)

```
plt.title("Correlation Between Sales Metrics")  
plt.show()
```

Important

Correlation does not prove causation.

For example, a relationship between quantity and profit does not necessarily mean that increasing quantity will always increase profit.

30. SQL DATABASE DESIGN

Load the cleaned dataset into MySQL.

30.1 Data Flow

Raw CSV

↓

Python / Pandas

↓

Data Cleaning

↓

Cleaned Dataset

↓

MySQL

↓

SQL Analysis

↓

Power BI

30.2 Example Table

```
CREATE TABLE ecommerce_sales (  
    Order_Date DATE,  
    Product_Name VARCHAR(255),  
    Category VARCHAR(100),  
    Region VARCHAR(100),  
    Quantity INT,  
    Sales DECIMAL(12,2),  
    Profit DECIMAL(12,2)  
);
```

Use appropriate data types and constraints based on the actual dataset.

31. SQL BUSINESS ANALYSIS

31.1 Total Sales

```
SELECT
    SUM(Sales) AS Total_Sales
FROM ecommerce_sales;
```

31.2 Total Profit

```
SELECT
    SUM(Profit) AS Total_Profit
FROM ecommerce_sales;
```

31.3 Sales by Category

```
SELECT
    Category,
    SUM(Sales) AS Total_Sales
FROM ecommerce_sales
GROUP BY Category
ORDER BY Total_Sales DESC;
```

31.4 Profit by Region

```
SELECT
    Region,
    SUM(Profit) AS Total_Profit
FROM ecommerce_sales
GROUP BY Region
ORDER BY Total_Profit DESC;
```

32. SQL — MONTHLY SALES ANALYSIS

```
SELECT
    YEAR(Order_Date) AS Sales_Year,
    MONTH(Order_Date) AS Sales_Month,
    SUM(Sales) AS Total_Sales
FROM ecommerce_sales
GROUP BY
```

```

YEAR(Order_Date),
MONTH(Order_Date)
ORDER BY
Sales_Year,
Sales_Month;

```

This helps identify monthly sales trends.

33. SQL — HIGH SALES, LOW PROFIT

First aggregate by product.

```

WITH product_summary AS (
  SELECT
    Product_Name,
    SUM(Sales) AS Total_Sales,
    SUM(Profit) AS Total_Profit
  FROM ecommerce_sales
  GROUP BY Product_Name
)
SELECT
  Product_Name,
  Total_Sales,
  Total_Profit,
  (Total_Profit / NULLIF(Total_Sales, 0)) * 100
  AS Profit_Margin
FROM product_summary
ORDER BY Total_Sales DESC;

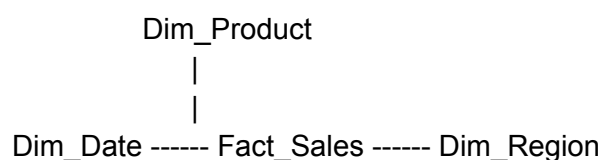
```

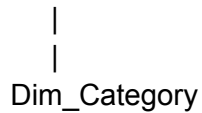
For a complete high-sales, low-profit analysis, define a threshold and filter products accordingly.

34. POWER BI DATA MODEL

Since this dataset may contain a single sales table, you can begin with a simple model.

For a more structured Power BI project, create a star schema.





Fact Table

Fact_Sales

Contains:

- Order Date
- Product
- Category
- Region
- Quantity
- Sales
- Profit

Dimension Tables

- Dim_Date
- Dim_Product
- Dim_Region
- Dim_Category

The exact design depends on the available fields.

35. POWER BI DATA PREPARATION

Use Power Query to:

- Change data types
 - Remove invalid rows
 - Handle missing values
 - Rename columns
 - Create calculated columns
 - Standardize categories
 - Validate financial values
-

36. DAX MEASURES

36.1 Total Sales

Total Sales =
SUM(Fact_Sales[Sales])

36.2 Total Profit

Total Profit =
SUM(Fact_Sales[Profit])

36.3 Total Quantity

Total Quantity =
SUM(Fact_Sales[Quantity])

36.4 Total Transactions

If each row represents one transaction:

Total Transactions =
COUNTROWS(Fact_Sales)

If there is a unique order ID, use the appropriate distinct count.

36.5 Overall Profit Margin

Profit Margin =
DIVIDE(
 [Total Profit],
 [Total Sales],
 0
)

Format this measure as a percentage.

36.6 Average Transaction Value

Average Transaction Value =
DIVIDE(
 [Total Sales],
 [Total Transactions],
 0
)

37. POWER BI DASHBOARD DESIGN

Create separate pages to organize the analysis.

A recommended structure is **five dashboard pages**.

38. DASHBOARD PAGE 1 — EXECUTIVE OVERVIEW

38.1 KPI Cards

Display:

- Total Sales
- Total Profit
- Total Transactions
- Total Quantity
- Profit Margin
- Average Transaction Value

38.2 Charts

Include:

- Monthly sales trend
- Sales by category
- Sales by region
- Profit by category

38.3 Main Business Question

How is the company performing overall?

39. DASHBOARD PAGE 2 — SALES & TREND ANALYSIS

39.1 Visualizations

- Monthly sales trend
- Monthly profit trend
- Sales by month
- Profit by month
- Sales and profit comparison

39.2 Filters

- Year
- Month
- Category
- Region

39.3 Main Business Question

How are sales and profitability changing over time?

40. DASHBOARD PAGE 3 — PRODUCT PERFORMANCE

40.1 Visualizations

- Top 10 products by sales
- Top 10 products by profit
- Top 10 products by quantity
- Sales by category
- Profit by category
- Product sales vs profit scatter plot

40.2 Main Business Question

Which products and categories contribute most to business performance?

41. DASHBOARD PAGE 4 — REGIONAL ANALYSIS

41.1 Visualizations

- Sales by region
- Profit by region
- Quantity by region
- Category performance by region
- Regional sales comparison

41.2 Main Business Question

Which regions are performing best, and where are improvement opportunities?

42. DASHBOARD PAGE 5 — PROFITABILITY ANALYSIS

42.1 Visualizations

- Profit margin by category
- Profit margin by region
- High-sales, low-profit products
- Sales vs profit analysis
- Product profitability table

42.2 Main Business Question

Which products and regions generate revenue but require profitability improvement?

43. KEY PERFORMANCE INDICATORS — KPI SUMMARY

Your dashboard should answer these questions immediately.

KPI

Business Meaning

Total Sales	Overall revenue generated
Total Profit	Overall profit generated
Total Transactions	Number of recorded transactions
Total Quantity	Total units sold
Profit Margin	Percentage of sales retained as profit
Average Transaction Value	Average sales value per transaction

44. INSIGHT GENERATION

This is one of the most important stages.

A chart alone is not an insight.

44.1 Weak Insight

Technology has the highest sales.

44.2 Stronger Insight

Technology generated the highest sales among the three categories, contributing X% of total revenue. However, its profit margin was lower than Office Supplies, indicating a need to investigate product costs, pricing, or discounts.

The actual conclusion must be based on the calculated results.

45. HIGH SALES, LOW PROFIT — BUSINESS INTERPRETATION

Suppose the analysis identifies a product with:

High Sales
Low Profit Margin

Possible reasons may include:

- High product costs
- Heavy discounting
- High shipping costs
- Low selling price
- High return-related costs, if available

Recommended Investigation

1. Identify the product.
2. Calculate total sales.
3. Calculate total profit.
4. Calculate profit margin.
5. Compare with similar products.
6. Investigate cost and pricing information if available.
7. Recommend further review.

Do not claim a specific cause unless the dataset contains supporting information.

46. BUSINESS RECOMMENDATIONS

Recommendations should be based on actual findings.

46.1 Product Recommendations

- Focus on products with strong sales and healthy profit margins.
- Investigate high-sales, low-profit products.
- Review pricing and cost structures where appropriate.
- Evaluate consistently underperforming products.

46.2 Regional Recommendations

- Study successful strategies in high-performing regions.
- Investigate low-profit regions.
- Adapt product strategies to regional demand.
- Compare category performance across regions.

46.3 Profitability Recommendations

- Monitor profit margin alongside sales.
- Avoid focusing only on revenue growth.
- Review products with high sales but low profit.

- Investigate pricing and cost drivers before making changes.
-

47. VALIDATE THE ANALYSIS

Before presenting the project, compare results across tools.

Python
↓
SQL
↓
Power BI

Example:

Metric	Python	SQL	Power BI
Total Sales	X	X	X
Total Profit	X	X	X
Total Quantity	X	X	X

The results should match, provided that the same filters, rows, and aggregation logic are used.

48. PROJECT LIMITATIONS

Include the following limitations in your report where applicable.

Limitation 1

The dataset contains simulated historical sales records, so the results may not represent an actual company's performance.

Limitation 2

The dataset covers 3,500 records and may not represent all possible sales situations.

Limitation 3

Customer-level information is not available in the described dataset.

Limitation 4

The dataset may not contain detailed cost components, such as shipping costs or discounts.

Limitation 5

High sales do not necessarily mean high profitability.

Limitation 6

Historical trends do not guarantee future performance.

49. FINAL PROJECT REPORT STRUCTURE

Your final documentation should contain:

1. Executive Summary
2. Project Introduction
3. Business Problem
4. Project Objectives
5. Dataset Description
6. Business Questions
7. Technology Stack
8. Data Dictionary
9. Data Loading
10. Data Profiling
11. Data Cleaning
12. Data Validation
13. Feature Engineering

14. Exploratory Data Analysis
 15. Overall Sales Analysis
 16. Monthly Trend Analysis
 17. Product Performance Analysis
 18. Category Analysis
 19. Regional Analysis
 20. Profitability Analysis
 21. High Sales, Low Profit Analysis
 22. SQL Analysis
 23. Power BI Data Model
 24. Dashboard Development
 25. KPI Creation
 26. Key Insights
 27. Business Recommendations
 28. Validation
 29. Limitations
 30. Conclusion
-

50. GITHUB PROJECT STRUCTURE

```
E-Commerce-Sales-Profitability-Analysis/  
|  
├── data/  
│   ├── raw/  
│   │   └── ecommerce_sales.csv  
│   └── processed/  
│       └── cleaned_ecommerce_sales.csv
```

```
├── notebooks/
│   ├── 01_data_understanding.ipynb
│   ├── 02_data_cleaning.ipynb
│   └── 03_exploratory_analysis.ipynb
├── sql/
│   └── analysis_queries.sql
├── powerbi/
│   └── ecommerce_sales_dashboard.pbix
├── reports/
│   └── project_report.pdf
├── screenshots/
│   ├── overview.png
│   ├── sales_trends.png
│   ├── product_analysis.png
│   ├── regional_analysis.png
│   └── profitability_analysis.png
├── README.md
└── requirements.txt
```

51. RESUME PROJECT DESCRIPTION

Once you complete the project and obtain the actual results, you can use a description like this:

E-Commerce Sales & Profitability Analysis | Python, SQL, Power BI

- Analyzed **3,500 historical e-commerce sales records** using Python, Pandas, and SQL to identify sales trends, product performance, and regional business patterns.
- Performed data cleaning, feature engineering, exploratory analysis, and profitability analysis to investigate high-sales, low-profit products across Technology, Furniture, and Office Supplies categories.
- Developed an interactive Power BI dashboard with sales, profit, quantity, regional performance, and product-level KPIs to support data-driven business recommendations.

Important: Add actual findings or measurable results after completing the analysis. Avoid claiming specific insights before calculating them.

52. INTERVIEW PROJECT EXPLANATION

You can explain the project using this structure:

52.1 Project Introduction

My project was an E-Commerce Sales and Profitability Analysis project based on 3,500 historical sales records from a simulated retail company.

52.2 Business Problem

The objective was to understand sales trends, regional performance, product performance, and profitability. The company wanted to identify products with high sales but low profit and improve its business strategies.

52.3 Data Cleaning

I loaded the dataset using Pandas and performed data profiling, missing-value analysis, duplicate checks, data-type validation, and numerical-value validation. I also converted the order date into a proper date format.

52.4 Analysis

I created monthly and yearly time-based features and performed exploratory analysis on sales, profit, quantity, products, categories, and regions. I also calculated profit margins to identify products with high revenue but comparatively low profitability.

52.5 SQL

I loaded the cleaned data into MySQL and used SQL queries with aggregation, grouping, and filtering to answer business questions related to sales, profit, products, and regions.

52.6 Power BI

Finally, I developed an interactive Power BI dashboard with separate pages for overall sales, trends, product performance, regional analysis, and profitability.

52.7 Outcome

The project helped identify business trends and profitability patterns and provided recommendations for further investigation into product pricing, costs, and regional performance.

53. COMPLETE END-TO-END PROCESS

STEP 1

Understand the Business Problem



STEP 2

Define Project Objectives



STEP 3

Understand Dataset



STEP 4

Identify Business Questions



STEP 5

Load Dataset Using Python



STEP 6

Check Rows and Columns



STEP 7

Inspect Data Types



STEP 8

Analyze Missing Values



STEP 9

Check Duplicate Records



STEP 10

Validate Numerical Values



STEP 11

Clean the Dataset



STEP 12

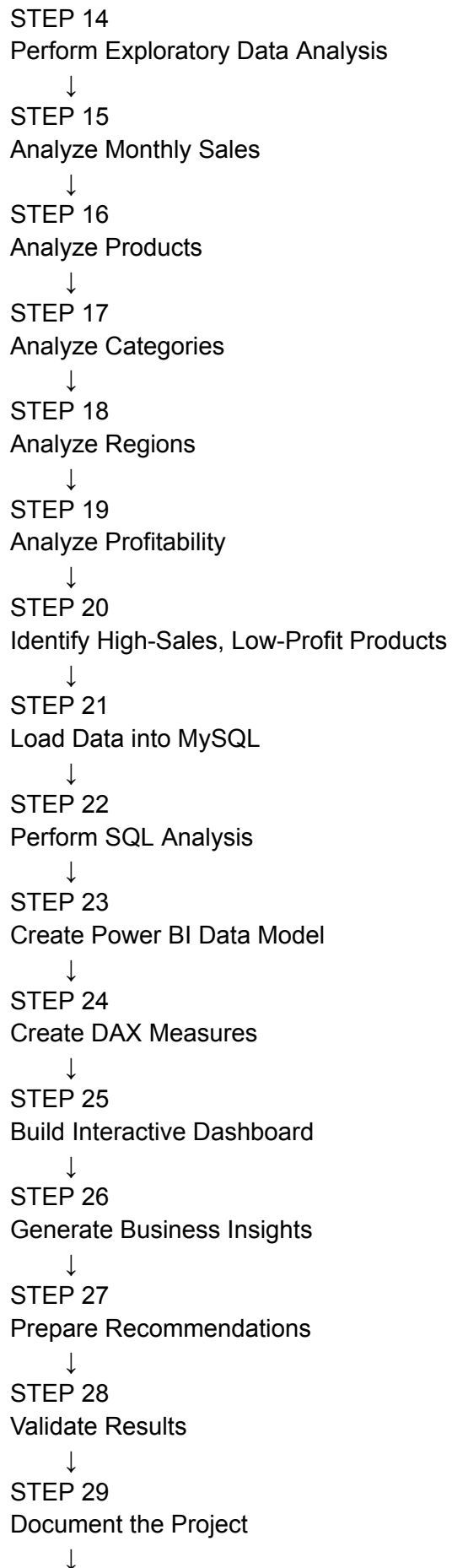
Create Date Features



STEP 13

Calculate Profit Margins





STEP 30
Upload to GitHub
↓
STEP 31
Add Project to Resume
↓
STEP 32
Prepare for Interviews

54. FINAL PROJECT POSITIONING

For your Data Analyst portfolio, present this project as:

E-Commerce Sales & Profitability Analysis

Skills Demonstrated:

Python
↓
Pandas
↓
Data Cleaning
↓
Feature Engineering
↓
EDA
↓
SQL
↓
Profitability Analysis
↓
Power BI
↓
DAX
↓
Dashboard Development
↓
Business Insights
↓
Recommendations

The most important business concept

High sales do not always mean high profit.

Your project should demonstrate that you can go beyond calculating revenue and identify where the company may need to investigate pricing, costs, product strategy, and regional performance.